

Thrombolytic Use in Acute Ischemic Stroke: Inclusion & Exclusion Criteria

Inclusion Criteria (for both Alteplase and Tenecteplase)

1. **Diagnosis of acute ischemic stroke** causing measurable neurologic deficit
2. **Age ≥ 18 years***
3. **Time from symptom onset to treatment:**
 - a. **Alteplase:** Within **4.5 hours**
 - b. **Tenecteplase:** Within **4.5 hours** (based on AHA/ASA and trial data)
4. **Non-contrast head CT or MRI ruling out hemorrhage**
5. **Neurologic deficit considered potentially disabling**

Exclusion Criteria

Category	Specific Exclusions
Hemorrhage risk	- Intracranial hemorrhage on imaging - Active internal bleeding - Recent (within 3 months) intracranial/spinal surgery, serious head trauma, or previous stroke - Platelets $< 100,000$ - INR > 1.7 or PT > 15 sec
Coagulopathy	- aPTT > 40 sec - LMWH within 24h at therapeutic dose - DOAC within 48h (unless normal renal function and labs)
BP/vascular	- SBP > 185 mmHg or DBP > 110 mmHg unresponsive to treatment - Known AVM, aneurysm, or neoplasm with high bleeding risk - Glucose < 50 mg/dL or > 400 mg/dL - Symptoms resolving rapidly
Other	- Seizure at stroke onset with postictal deficits only - Recent GI or GU hemorrhage (within 21 days) - Infective endocarditis or suspected aortic dissection - Pregnancy (relative exclusion)

Additional Exclusions Window – Alteplase

1. **Age >80 years**
2. **Use of oral anticoagulants**, regardless of INR
3. **NIH Stroke Scale (NIHSS) >25**
4. **History of both diabetes and prior stroke**

These were **protocol exclusions** in the ECASS III trial, but are now **relative exclusions** per 2019 AHA/ASA guidelines.

Tenecteplase-Specific Notes

- **Same core inclusion/exclusion criteria** apply as for alteplase.
- **Contraindications** are based on extrapolation from alteplase data and used in trials like **NOR-TEST**, **EXTEND-IA TNK**, and **ACT Trial**.

****Use in Pediatric Patients (<18 Years)***

While **alteplase is not FDA-approved for pediatric stroke**, it has been used **off-label** in select children with acute ischemic stroke. Use should be **restricted to specialized stroke centers** and guided by **neurology consultation**.

Key Data:

- **TIPS Trial (Thrombolysis in Pediatric Stroke, 2014):**
 - A multicenter trial evaluating IV alteplase in children aged 2–17 years.
 - **Terminated early due to enrollment challenges**, but showed feasibility and informed safety considerations.
 - Patients required confirmation of arterial ischemic stroke via imaging and dosing was age- and weight-adjusted.
- **Guidance from AHA 2019 Pediatric Stroke Scientific Statement:**
 - IV alteplase may be **considered in children ≥2 years** with:
 - Confirmed arterial ischemic stroke
 - No contraindications
 - Within 4.5 hours of onset
 - Care delivered at an institution with **pediatric stroke expertise**
 - **Dosing:** Same as adults (0.9 mg/kg, max 90 mg), though rarely used outside clinical trials

References

1. Powers WJ, Rabinstein AA, Ackerson T, et al. 2018 Guidelines for the Early Management of Patients With Acute Ischemic Stroke. *Stroke*. 2018;49(3):e46-e110. doi:10.1161/STR.0000000000000158
2. Powers WJ, Rabinstein AA, Ackerson T, et al. 2019 Update to the 2018 Guidelines for the Early Management of Acute Ischemic Stroke. *Stroke*. 2019;50(12):e344-e418. doi:10.1161/STR.0000000000000211
3. Hacke W, Kaste M, Bluhmki E, et al. Thrombolysis with alteplase 3 to 4.5 hours after acute ischemic stroke. *N Engl J Med*. 2008;359(13):1317-1329. doi:10.1056/NEJMoa0804656
4. Logallo N, Novotny V, Assmus J, et al. Tenecteplase versus alteplase for management of acute ischaemic stroke (NOR-TEST): a phase 3, randomised, open-label, blinded endpoint trial. *Lancet Neurol*. 2017;16(10):781-788. doi:10.1016/S1474-4422(17)30253-3
5. Campbell BCV, Mitchell PJ, Churilov L, et al. Tenecteplase versus alteplase before thrombectomy for ischemic stroke. *N Engl J Med*. 2018;378(17):1573-1582. doi:10.1056/NEJMoa1716405
6. Menon BK, Buck BH, Singh N, et al. Intravenous Tenecteplase Compared With Alteplase for Acute Ischemic Stroke in Canada (AcT): a pragmatic, multicenter, open-label, registry-linked, randomized, controlled, noninferiority trial. *Lancet*. 2022;400(10347):161-169. doi:10.1016/S0140-6736(22)00891-1